

Coding Challenge SQL

Crime Management

```
create database crimeManagement;
```

```
use crimeManagement;
```

```
CREATE TABLE Crime (  
    CrimeID INT PRIMARY KEY,  
    IncidentType VARCHAR(255),  
    IncidentDate DATE,  
    Location VARCHAR(255),  
    Description TEXT,  
    Status VARCHAR(20)  
);
```

```
desc crime;
```

```
select * from crime;
```

```
CREATE TABLE Victim (  
    VictimID INT PRIMARY KEY,  
    CrimeID INT,  
    Name VARCHAR(255),  
    ContactInfo VARCHAR(255),  
    Injuries VARCHAR(255),  
    FOREIGN KEY (CrimeID) REFERENCES Crime(CrimeID)  
);
```

```
select * from victim;
```

```
CREATE TABLE Suspect (  
  SuspectID INT PRIMARY KEY,  
  CrimeID INT,  
  Name VARCHAR(255),  
  Description TEXT,  
  CriminalHistory TEXT,  
  FOREIGN KEY (CrimeID) REFERENCES Crime(CrimeID)  
);  
  
select * from suspect;
```

```
INSERT INTO Crime (CrimeID, IncidentType, IncidentDate, Location, Description, Status)  
VALUES  
(1, 'Robbery', '2023-09-15', '123 Main St, Cityville', 'Armed robbery at a convenience store',  
'Open'),  
(2, 'Homicide', '2023-09-20', '456 Elm St, Townsville', 'Investigation into a murder case',  
'Under Investigation'),  
(3, 'Theft', '2023-09-10', '789 Oak St, Villagetown', 'Shoplifting incident at a mall', 'Closed'),  
(4, 'Assault', '2023-10-05', '321 Maple St, Metrocity', 'Physical assault reported in a park',  
'Open'),  
(5, 'Burglary', '2023-10-12', '654 Pine St, Smalltown', 'House break-in reported', 'Under  
Investigation'),  
(6, 'Fraud', '2023-10-18', '987 Birch St, Bigcity', 'Online banking fraud case', 'Closed');  
  
select * from crime;
```

	CrimeID	IncidentType	IncidentDate	Location	Description	Status
▶	1	Robbery	2023-09-15	123 Main St, Cityville	Armed robbery at a convenience store	Open
	2	Homicide	2023-09-20	456 Elm St, Townsville	Investigation into a murder case	Under Investigation
	3	Theft	2023-09-10	789 Oak St, Villagetown	Shoplifting incident at a mall	Closed
	4	Assault	2023-10-05	321 Maple St, Metrocity	Physical assault reported in a park	Open
	5	Burglary	2023-10-12	654 Pine St, Smalltown	House break-in reported	Under Investigation
	6	Fraud	2023-10-18	987 Birch St, Bigcity	Online banking fraud case	Closed
•	NULL	NULL	NULL	NULL	NULL	NULL

INSERT INTO Victim (VictimID, CrimeID, Name, ContactInfo, Injuries)

VALUES

(1, 1, 'John Doe', 'johndoe@example.com', 'Minor injuries'),
 (2, 2, 'Jane Smith', 'janesmith@example.com', 'Deceased'),
 (3, 2, 'Robert Brown', 'robertbrown@example.com', 'Gunshot wounds'),
 (4, 3, 'Alice Johnson', 'alicejohnson@example.com', 'None'),
 (5, 4, 'Michael Brown', 'michaelbrown@example.com', 'Bruises and cuts'),
 (6, 4, 'Emma Wilson', 'emmawilson@example.com', 'Broken arm'),
 (7, 5, 'Emily White', 'emilywhite@example.com', 'Shock and trauma'),
 (8, 6, 'David Green', 'davidgreen@example.com', 'Financial loss');

select * from victim;

	VictimID	CrimeID	Name	ContactInfo	Injuries
▶	1	1	John Doe	johndoe@example.com	Minor injuries
	2	2	Jane Smith	janesmith@example.com	Deceased
	3	2	Robert Brown	robertbrown@example.com	Gunshot wounds
	4	3	Alice Johnson	alicejohnson@example.com	None
	5	4	Michael Brown	michaelbrown@example.com	Bruises and cuts
	6	4	Emma Wilson	emmawilson@example.com	Broken arm
	7	5	Emily White	emilywhite@example.com	Shock and trauma
	8	6	David Green	davidgreen@example.com	Financial loss

INSERT INTO Suspect (SuspectID, CrimeID, Name, Description, CriminalHistory)

VALUES

(1, 1, 'Robber 1', 'Armed and masked robber', 'Previous robbery convictions'),
 (2, 2, 'Unknown', 'Investigation ongoing', NULL),
 (3, 3, 'Suspect 1', 'Shoplifting suspect', 'Prior shoplifting arrests'),
 (4, 4, 'Attacker 1', 'Wearing a red hoodie', 'History of violence'),
 (5, 5, 'Burglar 2', 'Broke in through a window', 'Prior burglary charges'),
 (6, 6, 'Scammer X', 'Conducted online scams', 'Previous fraud convictions');

select * from suspect;

	SuspectID	CrimeID	Name	Description	CriminalHistory
▶	1	1	Robber 1	Armed and masked robber	Previous robbery convictions
	2	2	Unknown	Investigation ongoing	NULL
	3	3	Suspect 1	Shoplifting suspect	Prior shoplifting arrests
	4	4	Attacker 1	Wearing a red hoodie	History of violence
	5	5	Burglar 2	Broke in through a window	Prior burglary charges
	6	6	Scammer X	Conducted online scams	Previous fraud convictions

-- 1. Select all open incidents

select * from crime; select * from crime where status='open';

	CrimeID	IncidentType	IncidentDate	Location	Description	Status
▶	1	Robbery	2023-09-15	123 Main St, Cityville	Armed robbery at a convenience store	Open
	4	Assault	2023-10-05	321 Maple St, Metrocity	Physical assault reported in a park	Open
*	NULL	NULL	NULL	NULL	NULL	NULL

-- 2. Find the total number of incidents

select count(*) as TotalIncidents from crime;

	TotalIncidents
▶	6

-- 3. List all unique incident types

select distinct(incidentType) from crime;

	incidentType
►	Robbery
	Homicide
	Theft
	Assault
	Burglary
	Fraud

-- 4. Retrieve incidents that occurred between '2023-09-01' and '2023-09-10'

select * from crime where IncidentDate between '2023-09-01' and '2023-09-10';

	CrimeID	IncidentType	IncidentDate	Location	Description	Status
►	3	Theft	2023-09-10	789 Oak St, Villagetown	Shoplifting incident at a mall	Closed
*	NULL	NULL	NULL	NULL	NULL	NULL

-- 5. List persons involved in incidents in descending order of age

alter table victim add age int;

alter table suspect add age int;

-- update the victim

update victim set age = 35 where victimid = 1;

update victim set age = 28 where victimid = 2;

update victim set age = 42 where victimid = 3;

update victim set age = 30 where victimid = 4;

update victim set age = 25 where victimid = 5;

update victim set age = 40 where victimid = 6;

update victim set age = 33 where victimid = 7;

update victim set age = 29 where victimid = 8;

select * from victim;

	VictimID	CrimeID	Name	ContactInfo	Injuries	age
▶	1	1	John Doe	johndoe@example.com	Minor injuries	35
	2	2	Jane Smith	janesmith@example.com	Deceased	28
	3	2	Robert Brown	robertbrown@example.com	Gunshot wounds	42
	4	3	Alice Johnson	alicejohnson@example.com	None	30
	5	4	Michael Brown	michaelbrown@example.com	Bruises and cuts	25
	6	4	Emma Wilson	emmawilson@example.com	Broken arm	40
	7	5	Emily White	emilywhite@example.com	Shock and trauma	33
	8	6	David Green	davidgreen@example.com	Financial loss	29
*	NULL	NULL	NULL	NULL	NULL	NULL

-- update the suspect

update suspect set age = 38 where suspectid = 1;

update suspect set age = 45 where suspectid = 2;

update suspect set age = 27 where suspectid = 3;

update suspect set age = 50 where suspectid = 4;

update suspect set age = 32 where suspectid = 5;

update suspect set age = 41 where suspectid = 6;

select * from suspect;

	SuspectID	CrimeID	Name	Description	CriminalHistory	age
▶	1	1	Robber 1	Armed and masked robber	Previous robbery convictions	38
	2	2	Unknown	Investigation ongoing	NULL	45
	3	3	Suspect 1	Shoplifting suspect	Prior shoplifting arrests	27
	4	4	Attacker 1	Wearing a red hoodie	History of violence	50
	5	5	Burglar 2	Broke in through a window	Prior burglary charges	32
	6	6	Scammer X	Conducted online scams	Previous fraud convictions	41
*	NULL	NULL	NULL	NULL	NULL	NULL

-- only victims

select name, age from victim order by age desc;

	name	age
▶	Robert Brown	42
	Emma Wilson	40
	John Doe	35
	Emily White	33
	Alice Johnson	30
	David Green	29
	Jane Smith	28
	Michael Brown	25

-- only suspect

select name, age from suspect order by age desc;

	name	age
▶	Attacker 1	50
	Unknown	45
	Scammer X	41
	Robber 1	38
	Burglar 2	32
	Suspect 1	27

-- both victim and suspect

select name,age,'victim' as role from victim

union

select name,age,'suspect' as role from suspect order by age desc;

	name	age	role
▶	Attacker 1	50	suspect
	Unknown	45	suspect
	Robert Brown	42	victim
	Scammer X	41	suspect
	Emma Wilson	40	victim
	Robber 1	38	suspect
	John Doe	35	victim
	Emily White	33	victim
	Burglar 2	32	suspect
	Alice Johnson	30	victim
	David Green	29	victim
	Jane Smith	28	victim
	Suspect 1	27	suspect
	Michael Brown	25	victim

-- 6. Find the average age of persons involved in incidents

```
select avg(age) as avg_age from ( select age from victim union all select age from suspect )as tot_person;
```

	avg_age
▶	35.3571

-- 7. List incident types and their counts, only for open cases.

```
select * from crime; select incidenttype,count(*) as incidentCount from crime where status='open' group by incidenttype;
```

	incidenttype	incidentCount
▶	Robbery	1
	Assault	1

-- 8. Find persons with names containing 'Doe'

```
select name from victim where name like '%Doe%';
```

	name
▶	John Doe

```
select name from suspect where name like '%Doe%';
```

	name

-- 9. Retrieve the names of persons involved in open cases and closed cases

```
select v.name , 'victim' as role , c.status from victim as v join crime as c on v.crimeid=c.crimeid where c.status='open' or c.status='closed'
```

```
union all
```

```
select s.name , 'suspect' as role,c.status from suspect as s join crime as c on
s.crimeid=c.crimeid where status='open' or c.status='closed';
```

	name	role	status
▶	John Doe	victim	Open
	Alice Johnson	victim	Closed
	Michael Brown	victim	Open
	Emma Wilson	victim	Open
	David Green	victim	Closed
	Robber 1	suspect	Open
	Suspect 1	suspect	Closed
	Attacker 1	suspect	Open
	Scammer X	suspect	Closed

-- 10. List incident types where there are persons(victim and suspect) aged 30 or 35 involved

```
select distinct incidenttype from crime as c join victim as v on c.crimeid=v.crimeid where
v.age in(30,35)
```

union

```
select distinct incidenttype from crime as c join suspect as s on c.crimeid=s.crimeid where
s.age in (30,35);
```

	incidenttype
▶	Robbery
	Theft

-- 11. Find persons involved in incidents of the same type as 'Robbery'

```
select v.name,'victim' as role, c.incidenttype
```

```
from victim v
```

```
join crime c
```

```
on v.crimeid = c.crimeid
```

```
where c.incidenttype='Robbery'
```

```
union all
```

```

select s.name,'suspect' as role, c.incidenttype
from suspect s
join crime c
on s.crimeid = c.crimeid
where c.incidenttype='Robbery';

```

	name	role	incidenttype
▶	John Doe	victim	Robbery
	Robber 1	suspect	Robbery

-- 12. List incident types with more than one open case

```

select incidenttype,count() as open
from crime
where status='open'
group by incidenttype
having count(>1;

```

incidenttype	open
--------------	------

-- 13. List all incidents with suspects whose names also appear as victims in other incidents

```

select c.crimeid,c.incidenttype,s.name
from suspect as s
join crime as c
on s.crimeid=c.crimeid
where s.name in(select v.name from victim as v where v.crimeid<>s.crimeid);

```

insert into victim (victimid, crimeid, name, contactinfo, injuries, age) values (9, 4, 'Robber 1', 'robber1@example.com', 'Minor injuries', 38);

Before insert:

	crimeid	incidenttype	name
--	---------	--------------	------

After insert:

	crimeid	incidenttype	name
▶	1	Robbery	Robber 1

-- 14.. Retrieve all incidents along with victim and suspect details

select c.crimeid, c.incidenttype, c.incidentdate, c.location, v.name as victim_name,
v.contactinfo, v.injuries, s.name as suspect_name, s.description as suspect_description
from crime c

left join victim v

on c.crimeid = v.crimeid

left join suspect s

on c.crimeid = s.crimeid

order by c.crimeid;

	crimeid	incidenttype	incidentdate	location	victim_name	contactinfo	injuries	suspect_name	suspect_description
▶	1	Robbery	2023-09-15	123 Main St, Cityville	John Doe	johndoe@example.com	Minor injuries	Robber 1	Armed and masked robber
	2	Homicide	2023-09-20	456 Elm St, Townsville	Jane Smith	janesmith@example.com	Deceased	Unknown	Investigation ongoing
	2	Homicide	2023-09-20	456 Elm St, Townsville	Robert Brown	robertbrown@example.com	Gunshot wounds	Unknown	Investigation ongoing
	3	Theft	2023-09-10	789 Oak St, Villagetown	Alice Johnson	alicejohnson@example.com	None	Suspect 1	Shoplifting suspect
	4	Assault	2023-10-05	321 Maple St, Metrocity	Michael Brown	michaelbrown@example.com	Bruises and cuts	Attacker 1	Wearing a red hoodie
	4	Assault	2023-10-05	321 Maple St, Metrocity	Emma Wilson	emmawilson@example.com	Broken arm	Attacker 1	Wearing a red hoodie
	4	Assault	2023-10-05	321 Maple St, Metrocity	Robber 1	robber1@example.com	Minor injuries	Attacker 1	Wearing a red hoodie
	5	Burglary	2023-10-12	654 Pine St, Smalltown	Emily White	emilywhite@example.com	Shock and trauma	Burglar 2	Broke in through a window
	6	Fraud	2023-10-18	987 Birch St, Bigcity	David Green	davidgreen@example.com	Financial loss	Scammer X	Conducted online scams

-- 15.Find incidents where the suspect is older than any victim

```
select c.crimeid,c.incidenttype,s.name as suspect_name, s.age as suspect_age,v.name
as victim_name, v.age as victim_age
```

```
from crime c
```

```
join suspect s
```

```
on c.crimeid = s.crimeid
```

```
join victim v
```

```
on c.crimeid = v.crimeid
```

```
where s.age > all (select age from victim where crimeid = c.crimeid);
```

	crimeid	incidenttype	suspect_name	suspect_age	victim_name	victim_age
▶	1	Robbery	Robber 1	38	John Doe	35
	2	Homicide	Unknown	45	Jane Smith	28
	2	Homicide	Unknown	45	Robert Brown	42
	4	Assault	Attacker 1	50	Michael Brown	25
	4	Assault	Attacker 1	50	Emma Wilson	40
	4	Assault	Attacker 1	50	Robber 1	38
	6	Fraud	Scammer X	41	David Green	29

```
-- 16. Find suspects involved in multiple incidents
```

```
select * from suspect; select name, count(distinct crimeid) as incident_count
```

```
from suspect
```

```
group by name
```

```
having count(distinct crimeid) > 1;
```

```
-- Inserting a record in suspect table insert into suspect (suspectid, crimeid, name,
description, criminalhistory, age) values (7, 2, 'Robber 1', 'Repeat offender', 'Prior robbery
cases', 38);
```

Before insert:

	name	incident_count
--	------	----------------

After insert:

	name	incident_count
▶	Robber 1	2

-- 17. List incidents with no suspects involved

```
select c.crimeid, c.incidenttype, c.incidentdate, c.status , s.name as suspectName
```

```
from crime as c
```

```
left join suspect as s
```

```
on c.crimeid=s.crimeid
```

```
where s.Suspectid is null;
```

```
select * from crime;
```

```
insert into crime (crimeid, incidenttype, incidentdate, location, description, status)
```

```
values (7, 'Money Laundering', '2023-11-05', '555 Finance St, Capital City', 'Investigation  
into illegal money transfers', 'closed');
```

Before insert:

	crimeid	incidenttype	incidentdate	status	suspectName
--	---------	--------------	--------------	--------	-------------

After insert:

	crimeid	incidenttype	incidentdate	status	suspectName
▶	7	Money Laundering	2023-11-05	closed	NULL

-- 18. List all cases where at least one incident is of type 'Homicide' and all other incidents are of type 'Robbery'

```
select c1.crimeid,c1.incidenttype
```

```
from crime as c1
```

where c1.incidenttype='homicide'

and not exists(select 1 from crime as c2 where c2.crimeid=c1.crimeid and
c2.IncidentType<>'homicide' and c2.IncidentType<>'robbery');

	crimeid	incidenttype
►	2	Homicide

-- 19. Retrieve a list of all incidents and the associated suspects, showing suspects for
each incident, or 'No Suspect' if there are none

```
select c.crimeid,c.incidenttype,coalesce(s.name,'No Suspect') as suspectname  
from crime as c  
left join suspect as s  
on c.crimeid=s.crimeid;
```

	crimeid	incidenttype	suspectname
►	1	Robbery	Robber 1
	2	Homicide	Unknown
	2	Homicide	Robber 1
	3	Theft	Suspect 1
	4	Assault	Attacker 1
	5	Burglary	Burglar 2
	6	Fraud	Scammer X
	7	Money Laundering	No Suspect

-- 20. List all suspects who have been involved in incidents with incident types 'Robbery' or
'Assault'

```
select s.suspectid,s.name,c.incidenttype  
from suspect as s  
join crime as c  
on s.crimeid=c.crimeid  
where c.IncidentType='robbery' or c.IncidentType='assault';
```

	suspectid	name	incidenttype
▶	1	Robber 1	Robbery
	4	Attacker 1	Assault